

Smart Library Management System (SLMS) - Requirements Document

v1.0

1.0 System Overview

The Smart Library Management System (SLMS) is a comprehensive web-based application designed to streamline library operations. It manages physical book inventories, e-book licenses, user memberships, and the complex lending lifecycle. The primary goal is to digitize manual processes and improve the experience for both patrons and administrative staff.

2.0 Functional Requirements

2.1 REQ-001: The SLMS shall allow an authenticated Librarian actor to add, update, archive, and delete book records from the main database.

2.2 REQ-002: Registered members must be able to search the catalog using multi-parameter filters including Title, Author, Genre, and Publication Year.

2.3 REQ-003: The system should automatically generate and dispatch an email notification when a borrowed item is overdue. The notification must be processed very quickly to ensure the user is informed without delay.

2.4 REQ-004: Members shall be permitted to place a hold on a physical book that is currently checked out by another user. This feature is strictly governed by the permission matrix detailed in section 4.0 and the inventory limits outlined in REQ-001.

3.0 Non-Functional Requirements

3.1 SEC-001 (Security): All user passwords and sensitive personally identifiable information (PII) must be encrypted using the AES-256 standard at rest.

3.2 PERF-001 (Performance): The catalog search engine should return query results seamlessly and handle high traffic gracefully during peak hours.

4.0 Role Authentication and Permissions

Role	View Catalog	Manage Users	Borrow Books	Override Fines
Guest	Yes	No	No	No
Memb	Yes	No	Yes	No

Role	View Catalog	Manage Users	Borrow Books	Override Fines
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Librarian	Yes	Yes	Yes	Yes

5.0 Technical Constraints

5.1 TECH-001: The backend architecture must be implemented using Python and rely on a PostgreSQL relational database.

5.2 TECH-002: The frontend user interface shall be responsive and function perfectly on all modern devices and internet connections.